



JAIPUR NATIONAL UNIVERSITY

End Semester/Year Back Examination December -2025

Name of School:	School of Engineering & Technology
Name of Programme / Course:	B.Tech. (Computer Science)/B.Tech. CSE(Hons.) in Artificial Intelligence and Machine Learning/B.Tech. CSE(Hons.) in Cyber Security
Semester/Year:	III / II
Paper Code:	BTECSCO304T22
Title of Paper:	Analog Electronics Circuits
Time: 3 Hrs.	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

1. This question paper consists of two sections, Section-A and Section-B.
2. Section-A consists of seven questions and
3. Section-B consists of six questions.

Attempt any five questions from Section-A and any four questions from Section-B. Marks allotted are mentioned against each question

Question Number	SECTION - A	Marks Allotted	Bloom Level	Course Outcome
Q.1	State the function of the emitter, base, and collector in a transistor.	4	1	1
Q.2	Compare the input and output characteristics of CB, CE, and CC configurations.	4	4	1
Q.3	What is Miller's Theorem?	4	1	2
Q.4	What is meant by a small-signal transistor amplifier?	4	1	2
Q.5	Discuss how a FET operates as a voltage-controlled device.	4	2	3
Q.6	State Barkhausen's criterion for oscillation.	4	1	4

Q.7	Distinguish between open-loop and closed-loop configurations of an Op-Amp.	4	2	5
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SECTION - B

Q.1	Evaluate the advantages and disadvantages of self-bias (emitter bias) over fixed bias for maintaining bias stability.	10	5	1
Q.2	Compare the characteristics and performance parameters of common-base, common-emitter, and common-collector configurations.	10	4	2
Q.3	Justify the use of the Darlington pair in amplifier design for achieving high input impedance and current gain.	10	5	3
Q.4	Discuss the working principle, construction, and types of Metal Oxide Semiconductor Field Effect Transistor (MOSFET).	10	2	4
Q.5	Describe the function of LC oscillators such as Hartley and Colpitts oscillators.	10	3	2
Q.6	Describe with waveforms the operation of Class A, Class B, Class AB, and Class C amplifiers.	10	2	5